

AKHILESH KUMAR MISHRA

PYTHON FULL STACK DEVELOPER | DJANGO · FASTAPI · REST APIS

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EXPERIENCE

Software Development Engineer

CSC e-Governance Services India Ltd. (MeitY, Government of India)

Ministry of Electronics & Information Technology, Govt. of India

12/2023 – PRESENT | New Delhi, India

- Scaled AEPS, MATM and VATM orchestration to **500,000+ daily transactions** at a **99% success rate**, by making the **Django REST Framework** payment path idempotent with request keys, backoff retries and nightly reconciliation.
- Took transaction query p95 down **45%** across **PostgreSQL** and **MySQL**, by partitioning the hot tables and adding covering indexes chosen from the query plan.
- Cleared **35%** of the peak-traffic backlog, by moving slow work off the request path onto **Celery** workers with **Redis** broking and idempotent task retries.
- Delivered live transaction status in under **500ms**, by streaming updates over **FastAPI WebSockets** instead of client polling.
- Contained a bad release to one service instead of the whole portal for **2 million+ users**, by moving monolithic web services onto async **FastAPI** and **Django** microservices behind a versioned REST gateway.
- Cut authentication latency **50%**, by issuing **JWT** access/refresh pairs with **RBAC** claims and caching OTP verification state in **Redis**.

Software Engineer (Contract)

Independent Client Engagement

Real-Time Telemetry & Geofencing Platform

06/2023 – 11/2023 | Remote, India

- Built a workforce telemetry and geofencing system for **1,000+ field agents** end to end, on **FastAPI** and async **SQLAlchemy** with **Alembic** migrations.
- Held **99.9% uptime** while absorbing **10,000+ concurrent** location pings, by batching writes through a **Celery** and **Redis** ingestion pipeline.
- Brought dashboard load time down **30%**, by wiring the **React.js** telemetry view to **FastAPI WebSocket** endpoints instead of polling.

SELECTED PROJECTS

DigiPay API Platform

FastAPI service with versioned routers, correlation-ID request tracing, auth and rate-limit middleware and a documented deprecation policy, packaged with Docker and a CI pipeline. github.com/GitHub-akhilesh/Django_apis_digipay

STACK: Python / FastAPI / PostgreSQL / Docker / GitHub Actions

Polyglot Task Microservices

Django REST API, Flask utility service and FastAPI notification service running side by side under Docker Compose with Kubernetes manifests, fronted by a React web client. github.com/GitHub-akhilesh/To-do-list-app

STACK: Python / Django / Flask / FastAPI / Docker / Kubernetes

Facial Recognition Attendance System

Automated attendance capture in Python using OpenCV Haar Cascade and LBPH recognition, with MySQL persistence and Pandas-generated Excel reports. github.com/GitHub-akhilesh/automatic-attendance-through-face-detection

STACK: Python / OpenCV / Pandas / Tkinter / MySQL

Async Telemetry Ingestion & Reporting Service

FastAPI ingestion service with Celery and Redis task queues writing to partitioned PostgreSQL tables, plus Pandas reporting jobs over aggregated telemetry.

STACK: Python / FastAPI / Celery / Redis / PostgreSQL / Pandas

SUMMARY

500,000+ transactions a day for 2 million+ users on India national e-governance platforms, served by Django REST Framework and async FastAPI services. Took transaction query p95 down 45% by partitioning and indexing PostgreSQL, cleared 35% of the peak backlog onto Celery and Redis workers, and streams live status over WebSockets in under 500ms. Reads the query plan before adding the index.

EDUCATION

Bachelor of Technology (B.Tech.)

Kanpur Institute of Technology

2019 – 2023 | Kanpur, UP

SKILLS

CORE PYTHON

Python Django Django REST Framework FastAPI
Flask Celery SQLAlchemy Alembic asyncio

APIS & SECURITY

RESTful APIs OpenAPI/Swagger JWT OAuth 2.0
RBAC rate limiting idempotency keys
FastAPI WebSockets API versioning

DATA & MESSAGING

PostgreSQL MySQL Redis Celery queues
covering indexes partitioning query plan analysis

TESTING & DEVOPS

pytest Docker Kubernetes GitHub Actions CI/CD
Linux Postman Prometheus Grafana React.js
JavaScript (ES6+)

CERTIFICATIONS

Database Systems & SQL Certification
Data Structures & Algorithms Certification
IIT Kanpur Cyber Security Certification
Advanced MERN Stack Developer Certification

ACHIEVEMENTS

- First Class Honors, B.Tech. — top academic rank in cohort.
- HackerRank 4-Star Gold badge in SQL.